



AuE6610 – Advanced and Electrified Powertrain

Course Project: Modelling & Control of a Dual-Mode Series–Parallel Hybrid Powertrain for Passenger Vehicles

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1.0 INTRODUCTION:

The main objective of the project is modelling, simulation and comparison of a naturally aspirated engine, electric and series-parallel hybrid powertrain intended for passenger cars. This study revolves around the performance of the above powertrains at varying conditions, their efficiencies and the optimal operating modes. Their relevance is evaluated against the driver demands and the ability to deliver the same. The main deliverable of the project include implementation of a series-parallel hybrid architecture that can operate the combustion engine and electric motor in their highest efficiencies respectively.

With the help of regenerative braking, drive-mode selection, battery state-of-charge restrictions, and energy management, the control technique employs a rule-based strategy for the best mode selection between series and parallel operation.

2.0 VEHICLE APPLICATION SELECTION:

2.1 Description of the Vehicle Application

- What is the chosen vehicle type? (e.g., city bus, passenger EV, race car)
- Typical use case, mission, and operating environment

2.2 Key Operational Requirements

- Expected payload/passenger load
- Target performance (e.g., acceleration, top speed, gradeability)
- Duty cycle characteristics

2.3 HEV Sedan Passenger/Family car

We have a selected a Hybrid Electric Vehicle designed for daily/family use in urban and highway driving conditions. This segment is the largest in the current automotive markets as the trend is slowly shifting towards electrification and hybridisation. With New Car prices and

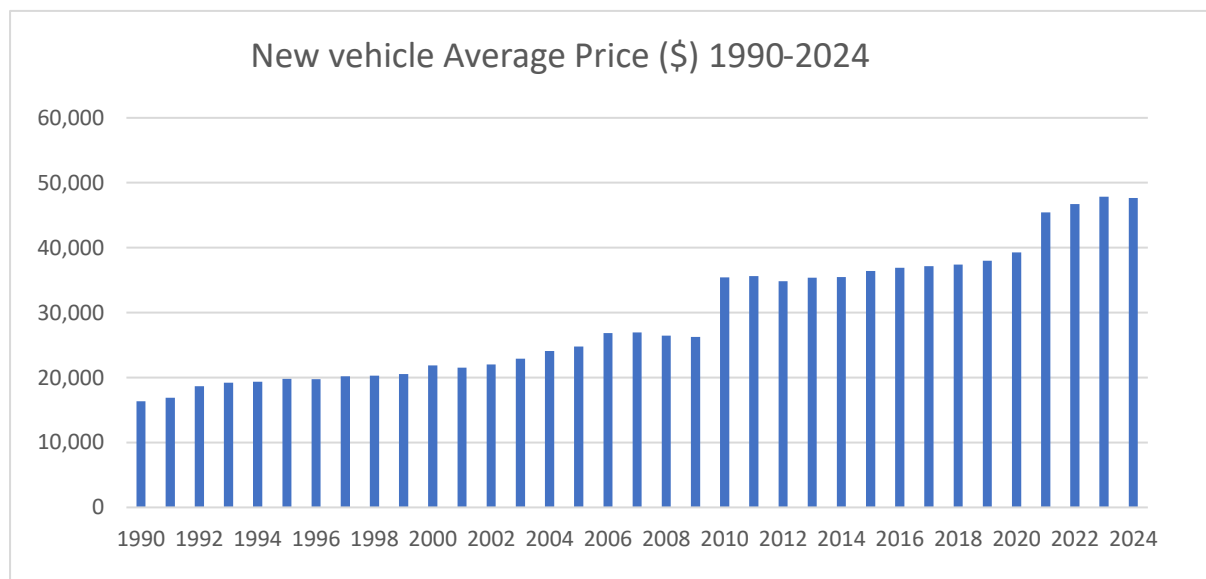


Figure 1: Sedan Vehicle over the years

Gasoline rates on a steep incline, the total cost of ownership for vehicles have skyrocketed in the past two decades.

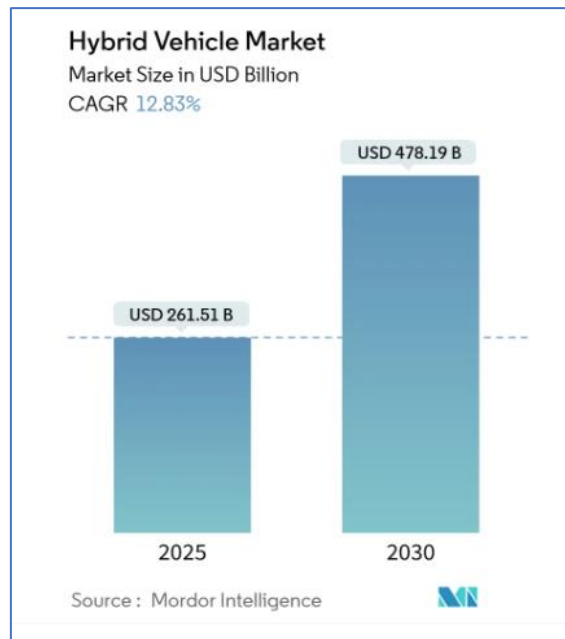


Figure 2: Hybrid Vehicle Market

Consumers are getting increasingly concerned about the rising fuel costs, contributing to significant growth in the EV industry. The Hybrid market currently sits at \$261 Billion and is expected to grow to almost \$480 Billion by the year 2030. This indicates strong demand and confidence in hybrid powertrains.

HEVs help reduce fuel consumption significantly compared to Combustion Engine vehicles. The addition of an electric motor assists the IC engine in improving fuel economy, with modern Hybrids reaching 50+ MPG in city and highway conditions as compared to 25-30 MPG for conventional vehicles. Based on consumption and driving patterns, this can save users up to \$2,000 dollars in fuel cost annually.

Another significant advantage of HEVs is their ability to recover energy through regenerative braking, further improving efficiency and reducing the wear and tear of brakes. Lastly, HEVs don't induce range anxiety in the consumers which is a major concern for EV owners. Buyers have the option to rely on either engine or motor for powering the vehicle all while enjoying an extended range and improved fuel economy.

3.0 MARKET ASSESSMENT:

3.1 Customer Requirements

Consumers look for the following requirements in a family Hybrid vehicle:-

- Driving range – 500-600 miles combined (suitable for long distances), eliminating range anxiety
- City and Highway Fuel Economy – 45-55 MPG
- Reliability – Drivers want reliability, which HEVs serve perfectly, by having the Engine as the main source of power. So even if the battery gets discharged or fails, the car can still rely on the engine to function, make it as reliable as CVs.
- Lower Maintenance costs – Having two sources of power prevents too much load on any one of them, reducing the wear and the need for frequent maintenance.

4.0 DRIVE CYCLE:

The drive cycle is collected with GPS logger app. The start point of the cycle was from CUICAR campus while destination was Clemson University Main campus.

- Distance covered: 37.8 miles
- Top Speed Achieved: 35.4 m/sec (79.9 mph)
- Travel Time: 55.9 minutes

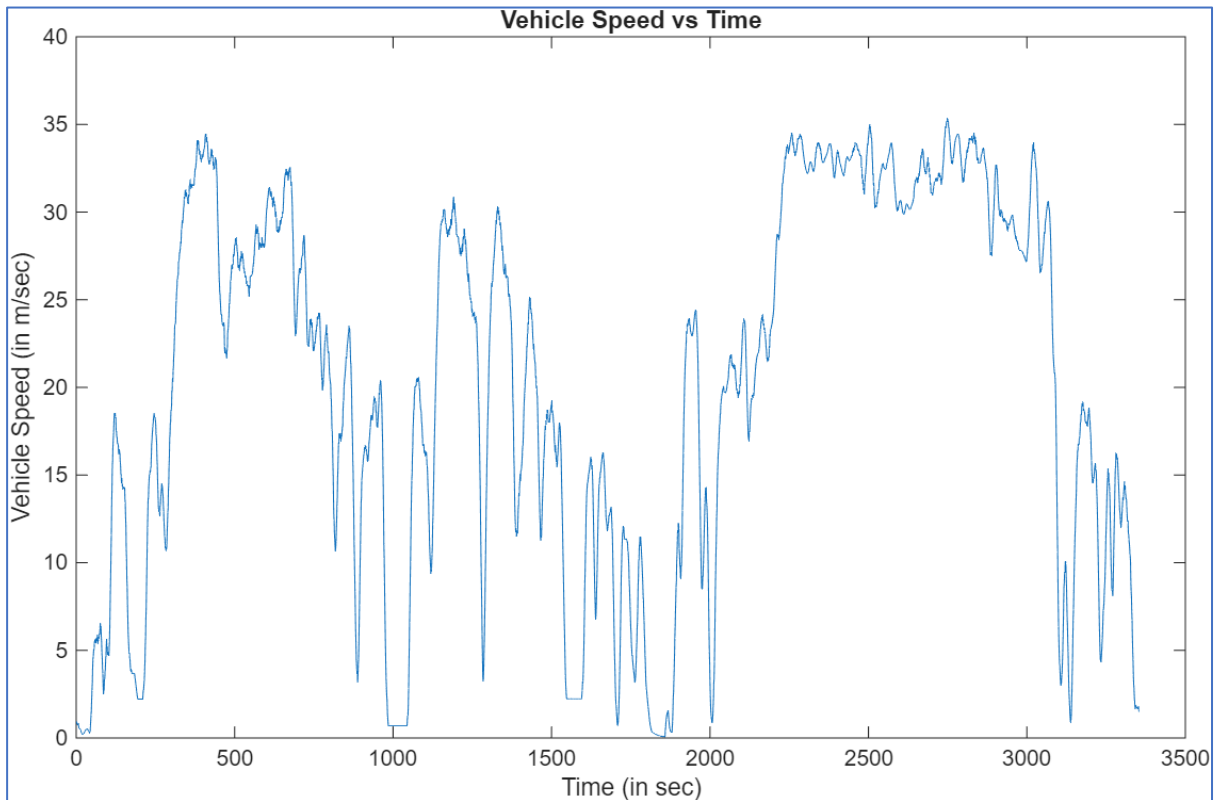


Figure 3: Drive Cycle (Velocity vs Time)

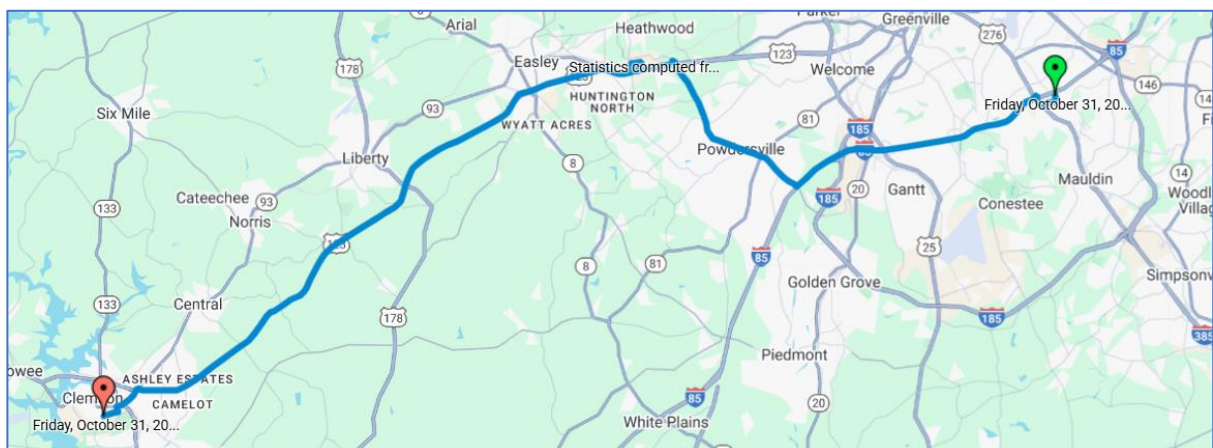


Figure 4: Drive cycle on Map

The peak power demand of this cycle is 52 kW. The energy consumed at wheels during this cycle is 40.405kWh.

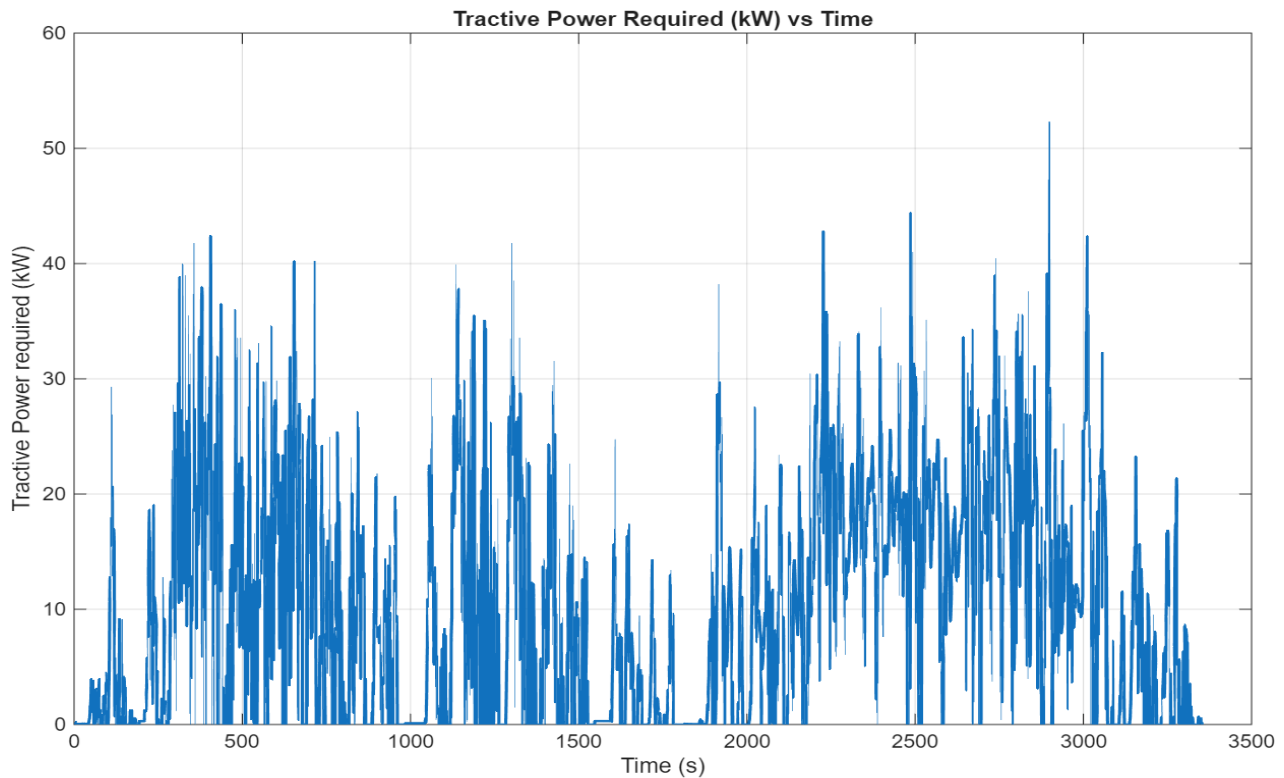


Figure 5: Tractive power demand of drive cycle

5.0 POWERTRAIN ARCHITECTURE:

Conventional, Electric and Series-Parallel Hybrid powertrains were simulated for the drive cycle. Their performance and impacts are discussed later. The vehicle assumptions for the powertrains are as follows:

- Curb Weight = 3337 kg
- Wheelbase = 108.3 inches
- Weight Distribution = 54:46
- CG height = 20 inches
- Cd Aero = 0.29
- Frontal area = 23.5 sq. f.
- Gear Ratio = [3.54, 1.961, 1.31, 0.97, 0.71, 0.62]
- Final Drive ratio = 4.21
- Wheel radius = 17 inches
- Rolling resistance coefficient = 0.01
- Tire friction coefficient = 0.85
- Transmission efficiency = 0.96
- Transmission inertia = 0.04 kg.m sq
- Final drive inertia = 0.02 kg.m sq
- Driveline efficiency = 0.95

5.1 CONVENTIONAL POWERTRAIN – NA ENGINE

Naturally aspirated 2-liter is selected based on power requirement. Toyota Corolla vehicle is referenced for benchmarking. The technical parameters of shortlisted engine are as follows:

- Bore: 82 mm
- Stroke: 82 mm
- Number of cylinders: 4

Volumetric and Thermal efficiency was taken from the homework data files.

Considering all the vehicle and engine parameters, The BSFC calculations are performed. The BSFC

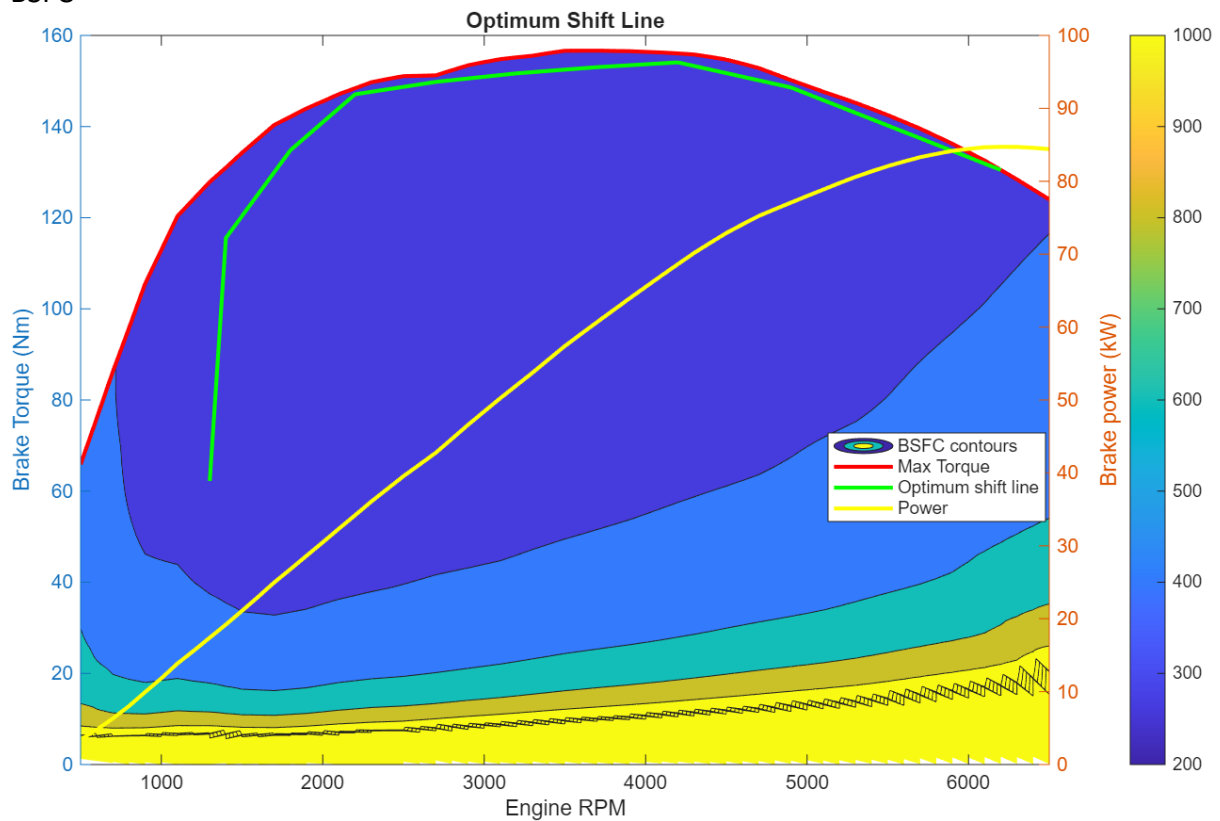


Figure 6: Engine BSFC map

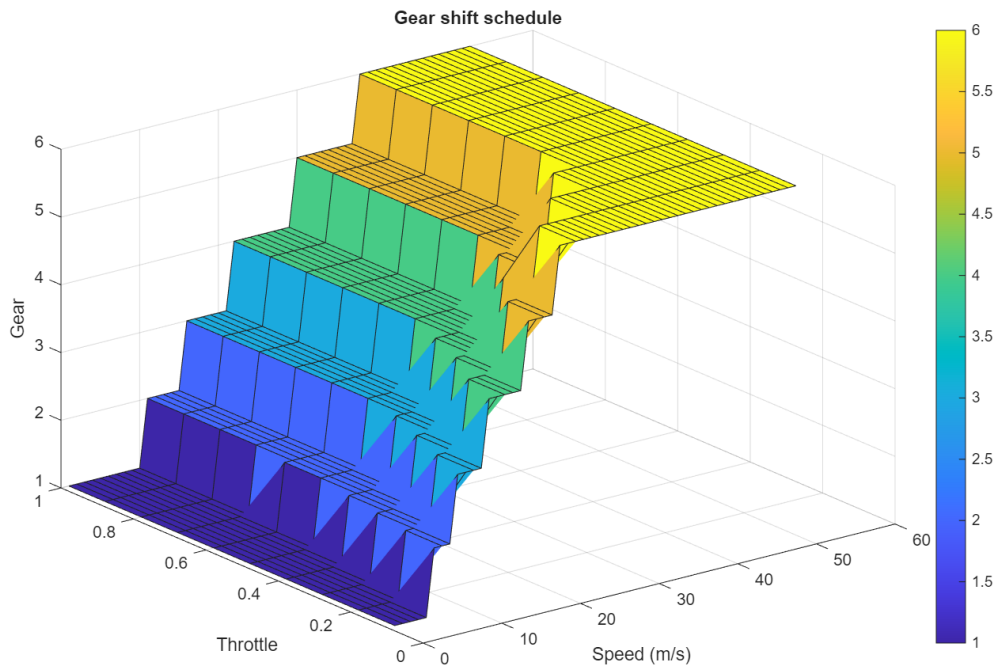


Figure 7: Gear Map

The Simulink model for engine and vehicle is constructed. PID controller is used to control pedal position.

Complete forward-facing vehicle longitudinal simulation model in Simulink that replicates vehicle behaviour under driving conditions using a BSFC-based gear-shift strategy. The model integrates driver behaviour, supervisory gear-selection logic, engine, drivetrain torque flow, wheel traction, and vehicle dynamics into a closed-loop feedback system.

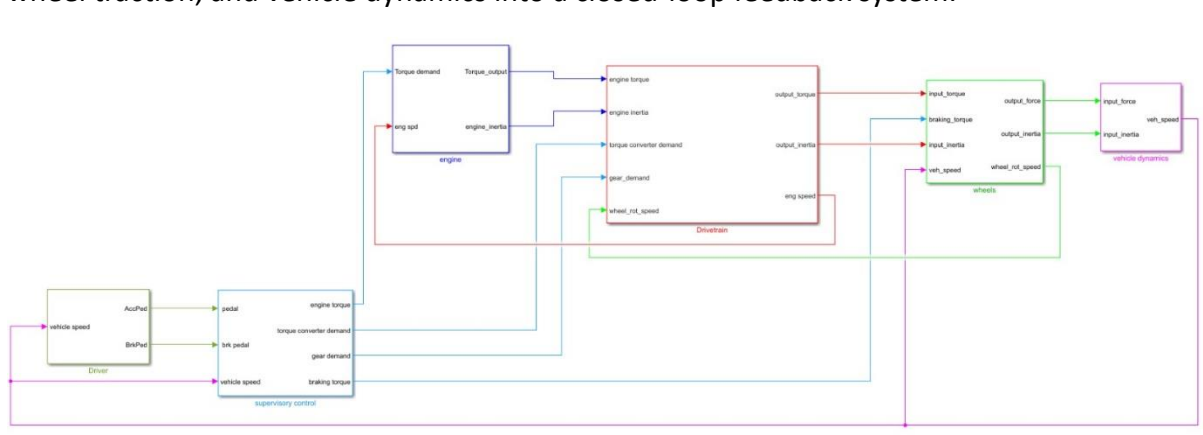


Figure 8: Simulink model of combustion vehicle

The model starts with a Driver block, which acts as a virtual representation of human driving input. Using a closed-loop feedback structure, the driver generates throttle (pedal) input proportional to the speed error. The drive cycle velocity is given as reference for PID to calculate error.

The signal later passes into the supervisory control layer, where gear management is handled. The 2-D lookup-based gear-shift map, which selects the most fuel-efficient gear based on vehicle speed and pedal position. Engine speed and torque demand are fed into a lookup-based brake-specific fuel consumption (BSFC) map. Using interpolation, the fuel rate is determined to calculate fuel economy.

Drivetrain receives input from engine block as well as from wheels. Torque converter, transmission, and final drive are included in this block. It converts torque from engine to wheels and speed from wheels to engine. The selected gear ratio multiplies engine torque, after applying drivetrain efficiency.

The wheels convert driving or braking torque into tractive force.

In vehicle dynamic block, all the forces acting on the vehicle are considered. Which includes tractive, Aero and rolling resistance. Forces results the acceleration of the vehicle. Later velocity of the vehicle is calculated by integrating the acceleration.

The major results of the simulation are:

- Engine operating points
- Fuel consumption for drive cycle
- 0-60 timings

Creation and simulation of a fully coupled vehicle simulation in Simulink, where engine performance, fuel consumption, wheel traction, and vehicle acceleration are coordinated through a BSFC-based gear-shift strategy were successfully achieved by simulation.

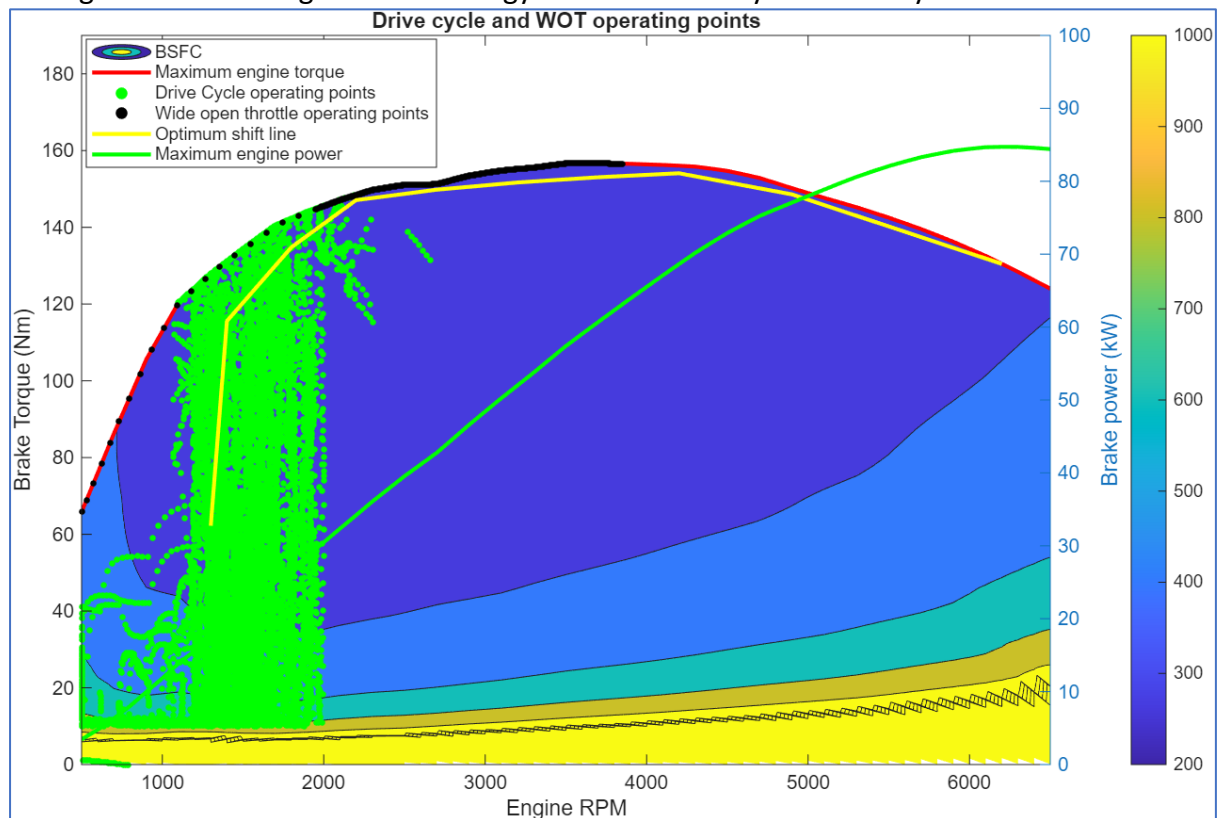


Figure 7: Engine operating points for drive cycle

The fuel economy for the considered vehicle powertrain is **43 MPG**.

5.2 ELECTRIC POWERTRAIN:

Based on the drive cycle's power demand the motor was shortlisted. Out of which the Borgwarner HVH250 standard is selected.

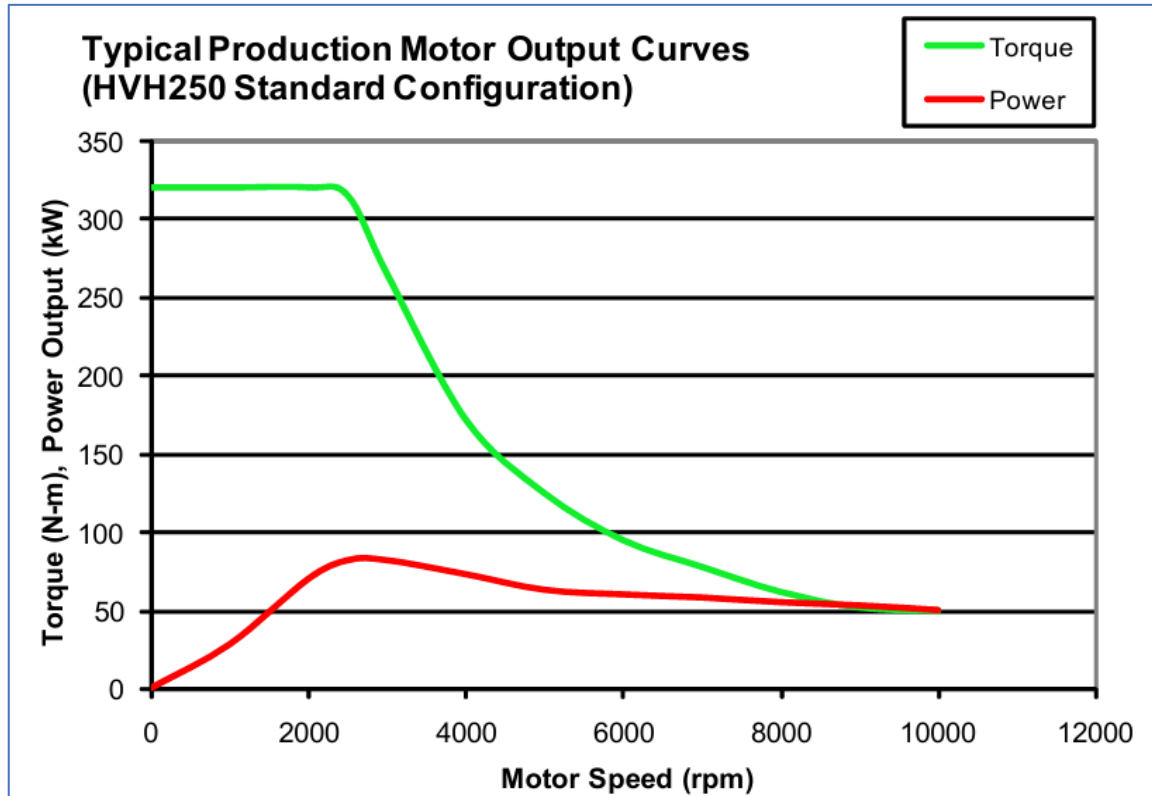


Figure 8: Motor Torque and Power vs RPM

The voltage rating of the motor is 320V. The high voltage system is then designed for this voltage. The cell selection is done on the basis of the energy density and the cycle life parameters. With the data available in homework 5 cells, battery sizing is done.

Nominal cell voltage : 3.6 V

Cell Capacity: 6.3 Ah

To achieve voltage of 320V, 90 cells are stacked in series. And to meet the energy requirement 20 such module then connected in parallel resulting in 90S20P configuration. The final battery specifications are as follows:

Battery Nominal Voltage: 324 V

Battery Capacity: 126 Ah

Battery size: 40.824 kWh

The above model is used to predict SOC, Terminal voltage and Power output for the given battery specifications given in HW5. In addition, there is also an electric motor and thermal control block added to the model (controls battery back temperature and coolant flowrates). The above model is simulated for the UDDS cycle over 7.5 miles.

This data is provided into Simulink to simulate the same vehicle with electric powertrain and motor as source.

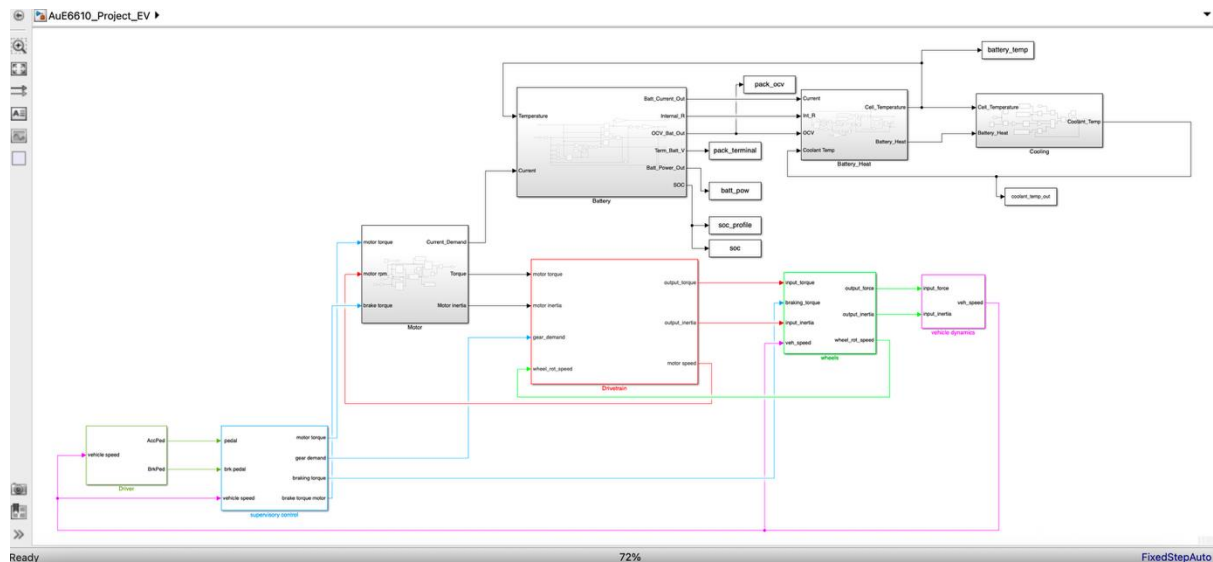


Figure 9: Simulink Model for EV

The input blocks receive data from the UDDS drive Cycle (Current demand). The motor block receives torque and rpm from supervisor control and transmission block. The transmission used for the vehicle is single stage fixed ratio transmission. The gear ratio is decided on the goal of maintaining motor operating in efficient zone and meeting wheel torque. The motor provides demanded electric power based on look up table. This allows to calculate the current demand for the cycle (Power/Voltage).

The flag TEMP signifies the battery pack temperature received from the thermal block. Battery block carries out for estimation of SOC. Current demand from the drive cycle is integrated over time and passed through gain block to convert it into Ampere – hour. This value is divided by the maximum capacity of the battery that is found through the 1-D lookup table using battery temperature and current demand. A block used to subtract this result from the initial SOC value (0.90), providing us with the instantaneous SOC for the drive cycle. Look up table gives resistance across the battery. Through the data provided for resistance against current demand and SOC, 1-D lookup tables are used to determine the resistance during charging and discharge cycle by providing them the current demand and SOC values. Furthermore, a switch is provided that will consider the battery to be in charging mode when the current supply is positive and in discharge mode when the current supply is negative. The open circuit voltage is simply obtained by using the lookup table and feeding it the current demand and state of charge values. This is the value obtained for a single cell. To plot OCV for battery, it is scaled by a gain of 1800 (100 cells in series, 20 parallel).

In this block, battery pack temperature is calculated based on the coolant flow, coolant temperature and the overall heat generation and dissipation carried out through the drive cycle.

The heat dissipated is obtained from the battery coolant block by modelling the mathematical calculation of $Q(\text{dissipated}) = h.A.\Delta T$, where delta T is the difference between battery pack and coolant temperatures.

Tdot is then obtained by the relation: $\Delta Q \div (m_{\text{cell}} * C_{p_{\text{cell}}})$. By setting the initial temperature of 25 degree C and integrating the output over time we get the instantaneous battery pack temperature.

As mentioned earlier, heat dissipated is calculated by using the formula: $Q = h.A.dT$. Since Aim is to maintain the battery temperature < 32 degrees C, a PID controller has been used to achieve this. The difference between battery temperature and constant limit (25 deg C) is

calculated. 25 degrees is set as the limit so that by the time the coolant flowrate is modified and cooling occurs, the temperature does not cross the 32 deg threshold. The PID output is fed to a saturation block, and the result is the coolant flowrate over time. The model resulted successful simulation with giving output data of SOC, OCV, Terminal Voltage, cell and coolant temperature.

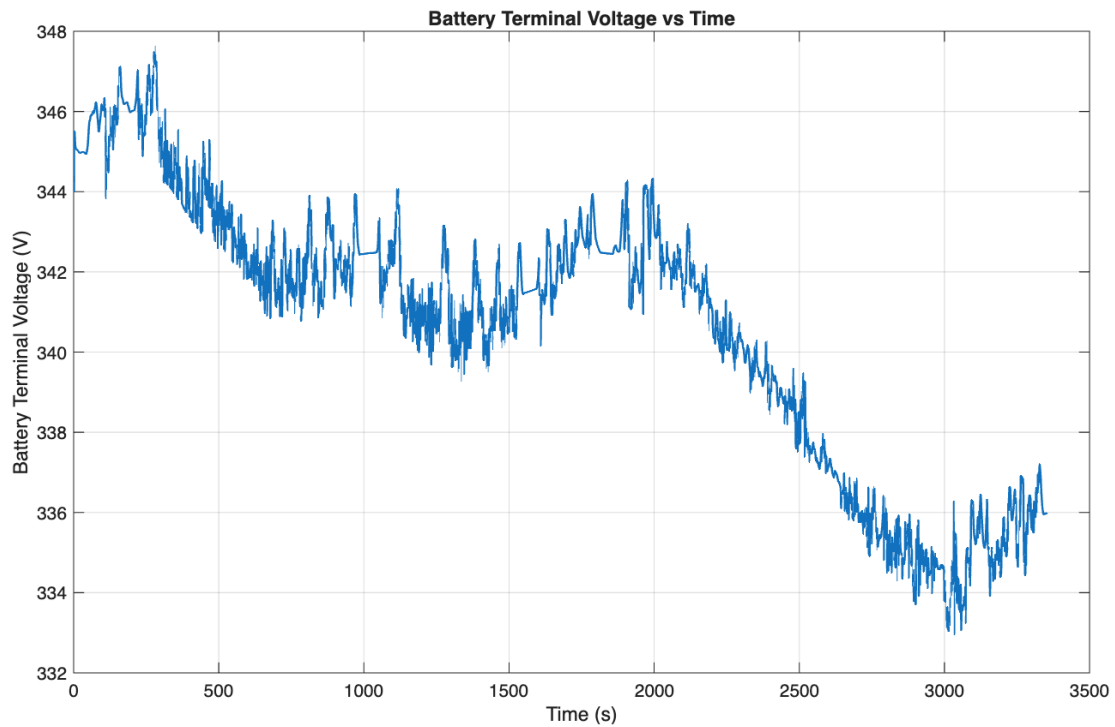


Figure 11: Battery terminal voltage

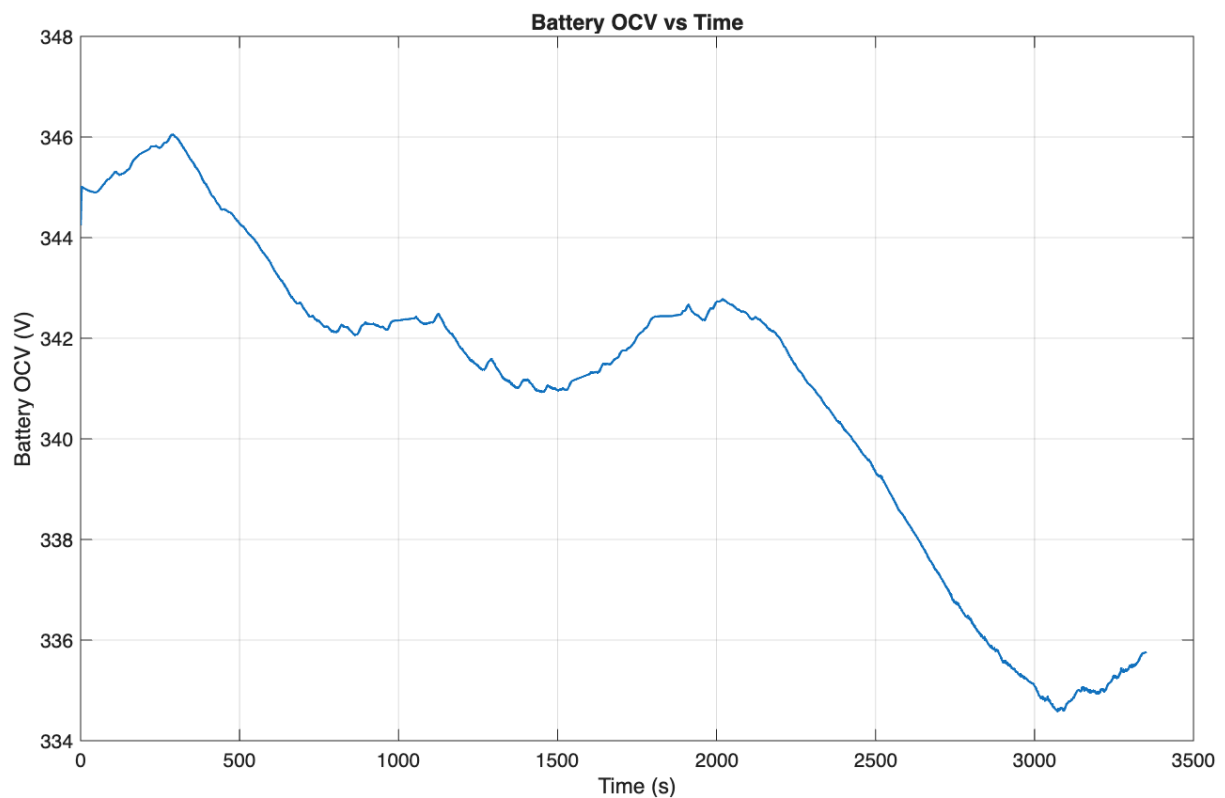


Figure 10: Battery Open cell voltage

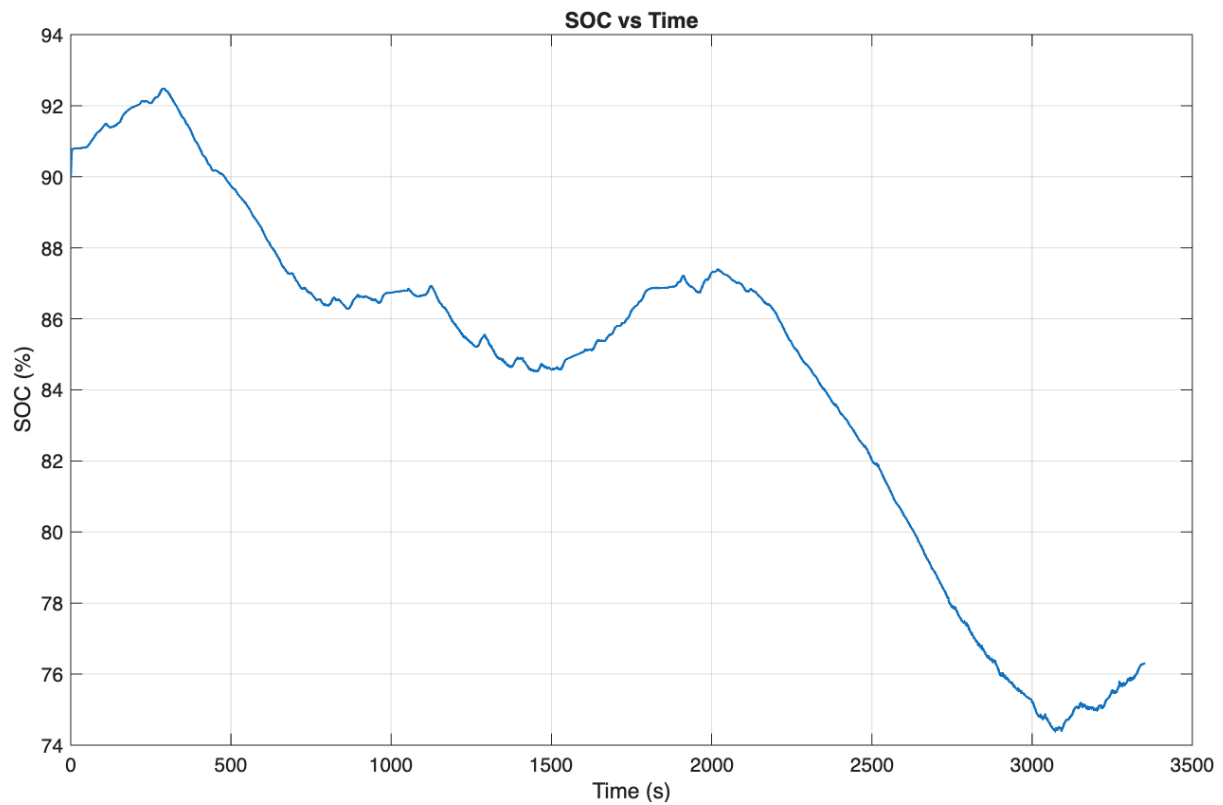


Figure 12: Battery SOC

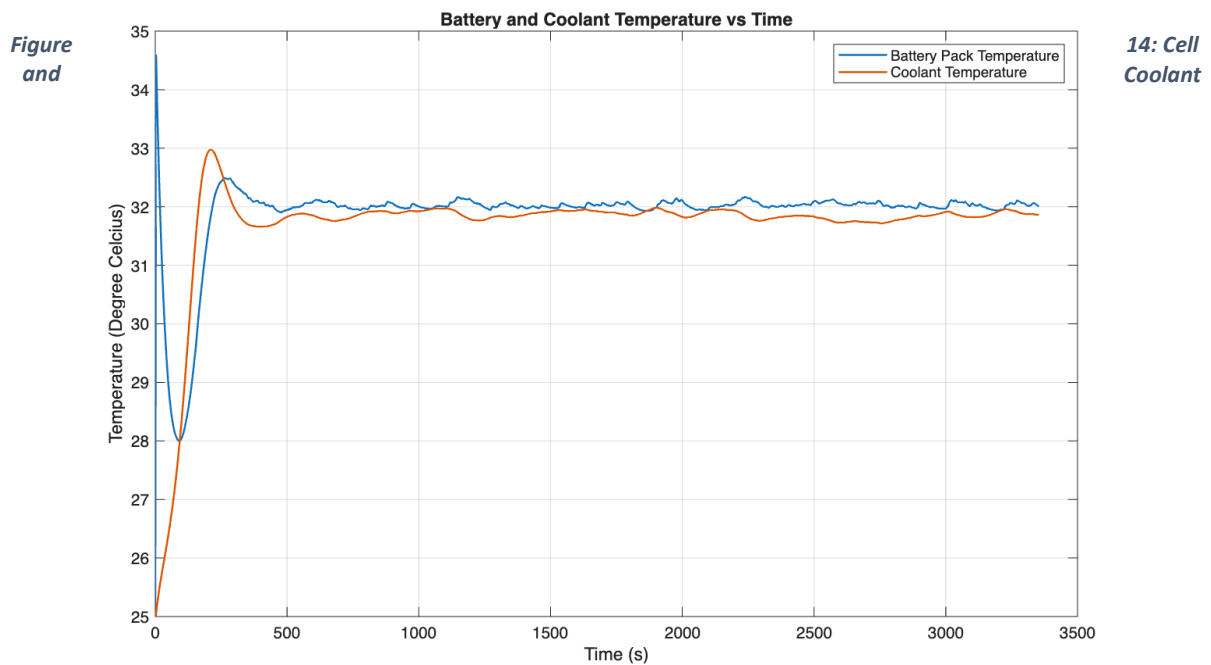


Figure 13 Battery and Coolant Temperature

Figure
and

14: Cell
Coolant

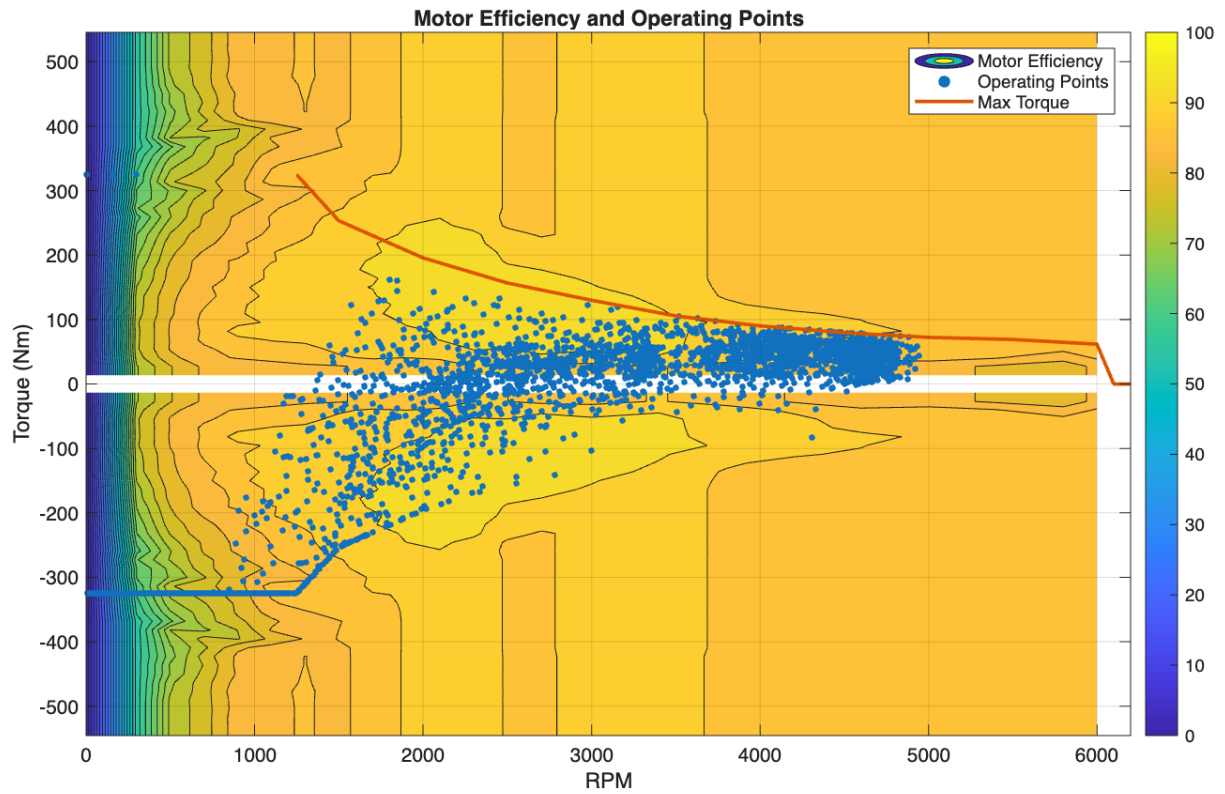


Figure 14: Motor operating points and efficiency plot

Based on SOC profile, the battery life is calculated. Assuming this cycle repeats for the rest of the battery life, the battery will survive for 186472.54 miles. The full electric range of the vehicle is 181.02 miles.

5.3 HYBRID POWERTRAIN:

For the HEV powertrain, we have decided on a series parallel hybrid. Considering the hybrid architecture, we have downsized the engine to meet the following specifications: -

- Bore: 75 mm
- Diameter: 87.9 mm
- 4 Cylinders, 1.5L
- Power: 57 kW

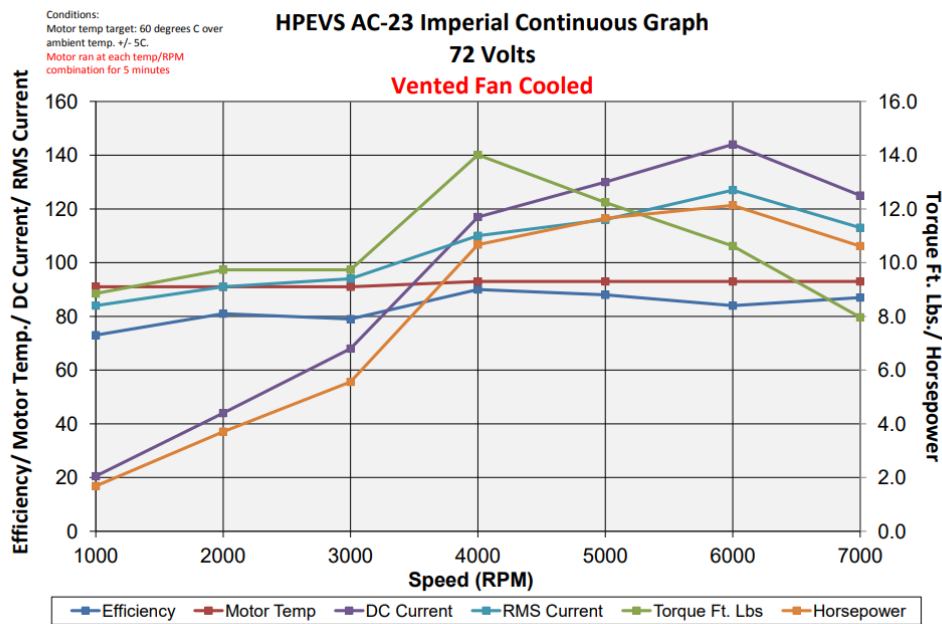


Figure 15: Hybrid motor characteristic

To compensate for and assist the downsized engine, we have chosen the HPEVS AC-23 Motor with the following specifications

- 72V, 300 Amps
- Max Torque: 60.33 Nm
- Max Power: 33 kW

This gives a combined Power output of 90 kW in Parallel mode Coming to the battery sizing, based on the motor selection, we decided to use the same cells used in our EV Battery, only this time to achieve the Voltage rating, we connected 20 cells in series and stacked 35 such modules in parallel (20s35p, achieving a low voltage high current system).

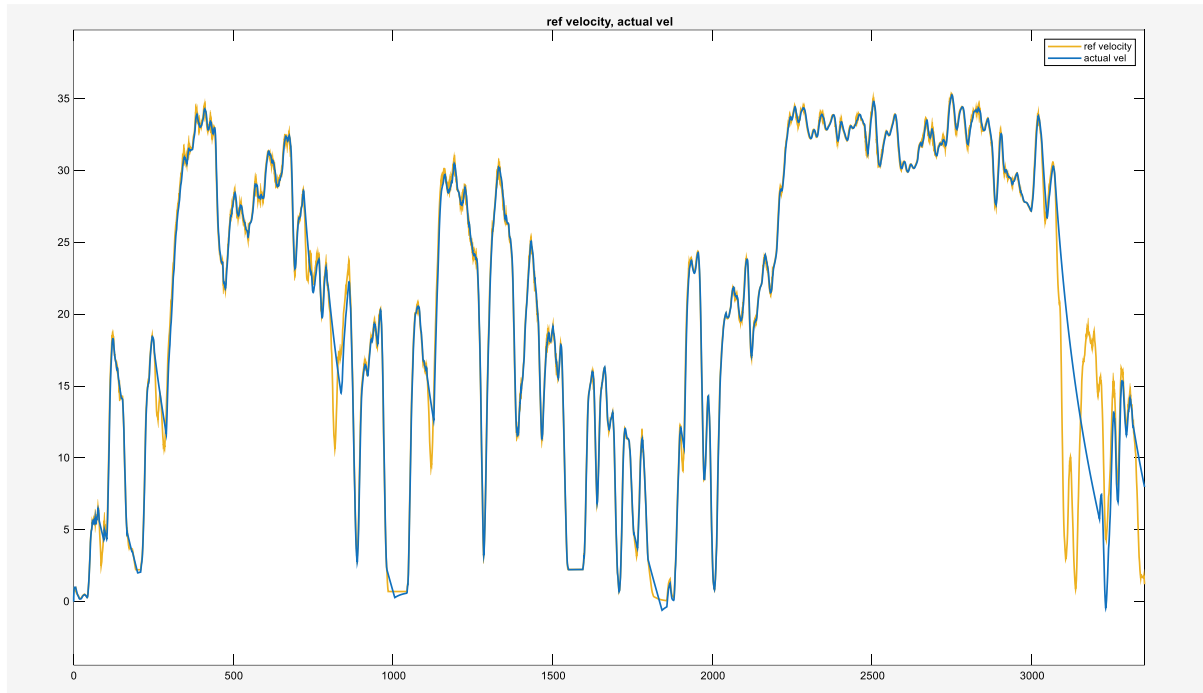
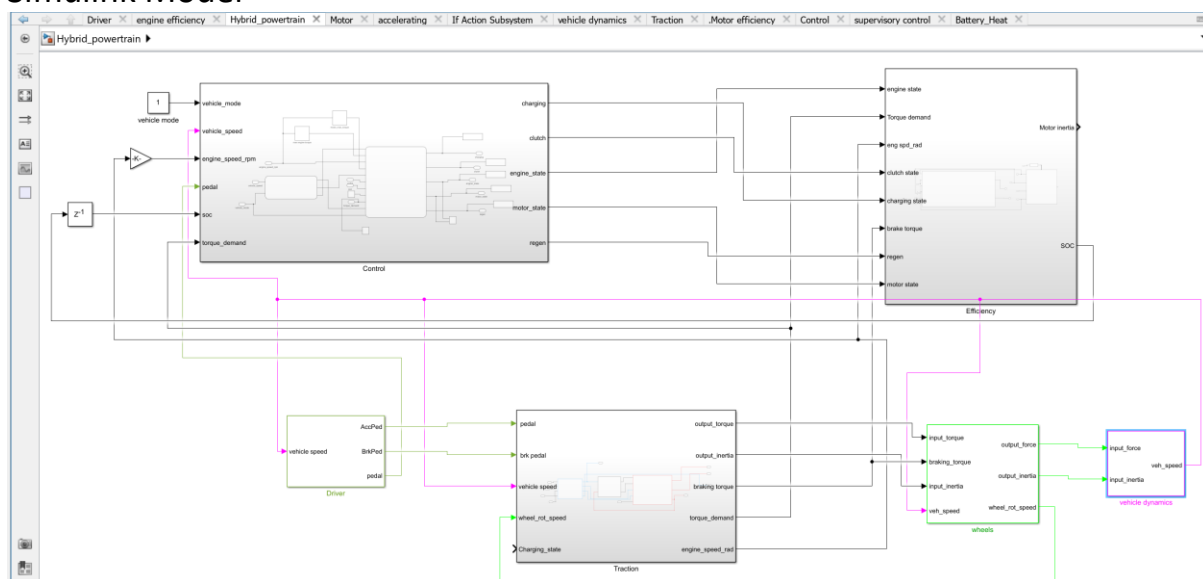


Figure 16 Ref vs Actual Velocity HEV

Simulink Model



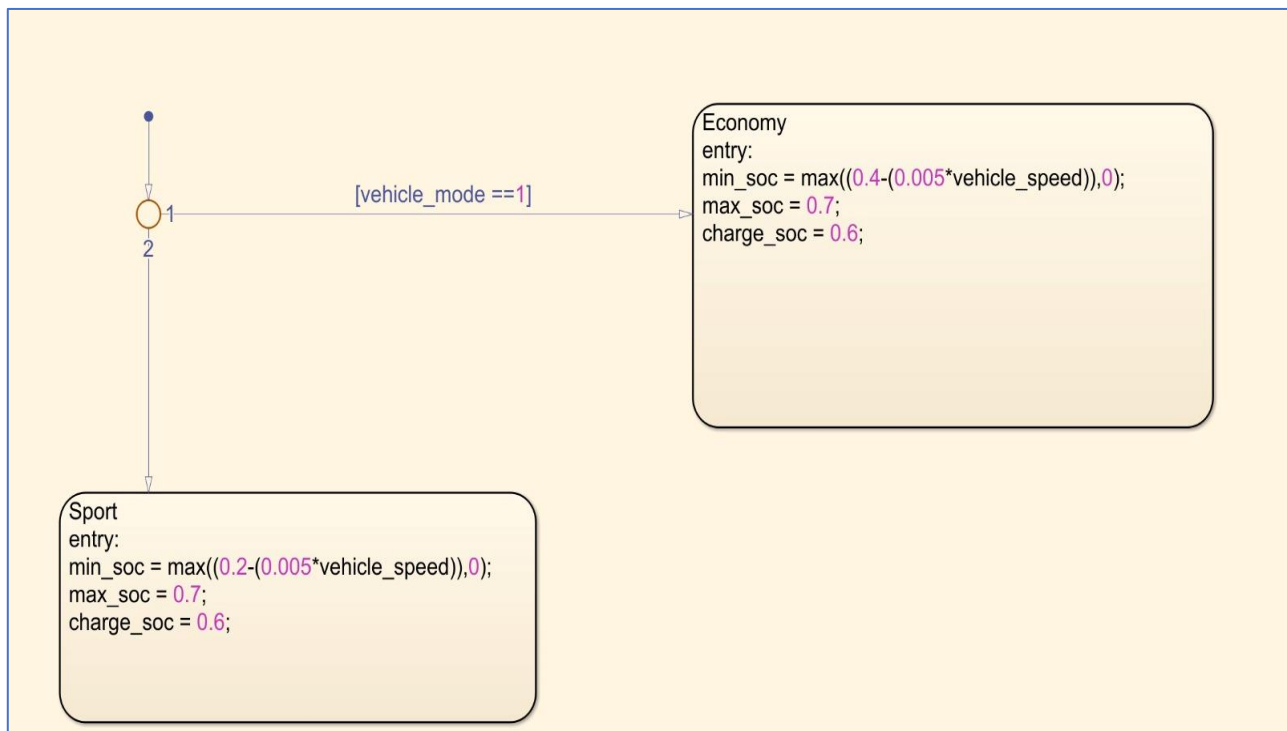
Flow of the model: -

The **Driver subsystem** calculates the pedal output between -1 to 1. This input is fed to the **Traction subsystem** which generates the torque demand and RPM of both engine and motor. This along with some other parameters is fed to the **Control subsystem** which controls the states of engine, motor, clutch and

regeneration subsystem. These states are used for calculating efficiency (MPG and SOC) in the **Efficiency subsystem**.

5.3.1 CONTROL STRATEGY: STATE OF CHARGE

SoC control block decides the Minimum and Maximum limit of SoC based on



the vehicle speed (more vehicle speed means higher chance of regaining the charge by regen) and vehicle mode (Economy and Sport). The SoC limits are adjusted dynamically throughout the simulation. These outputs are given to the state control blocks.

Formula used

SoC Minimum limit = Min_SoC(fixed for a vehicle mode) – 0.005*vehicle speed

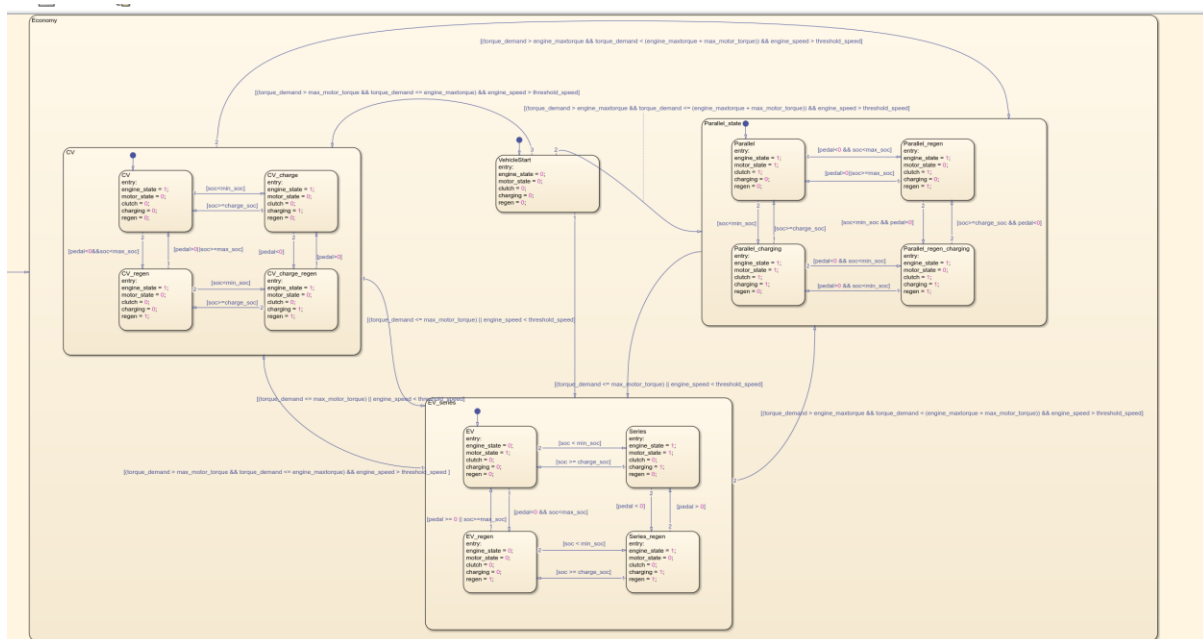
State Control blocks

State variables -

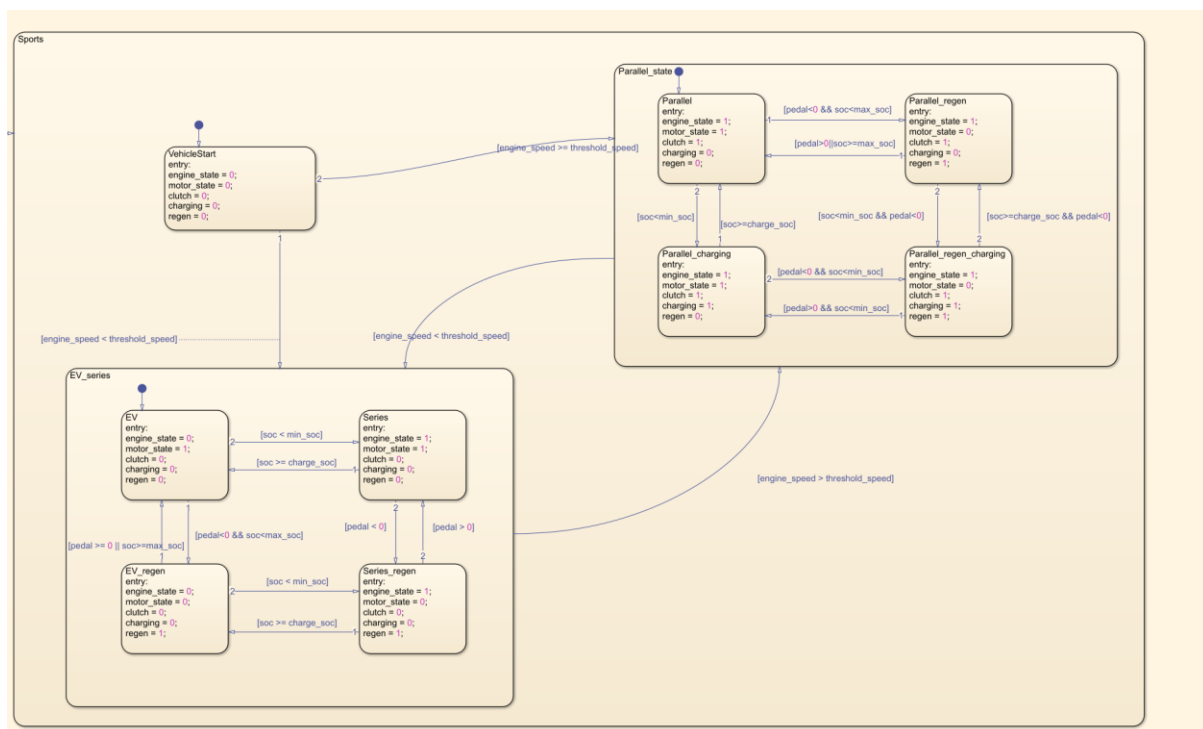
- 1) Engine state
- 2) Motor state
- 3) Clutch state
- 4) Regen state
- 5) Charging state (engine charging the battery)

States –

- 1) Economy mode
- 2) Sport mode



SPORTS MODE(vehicle mode = 2):

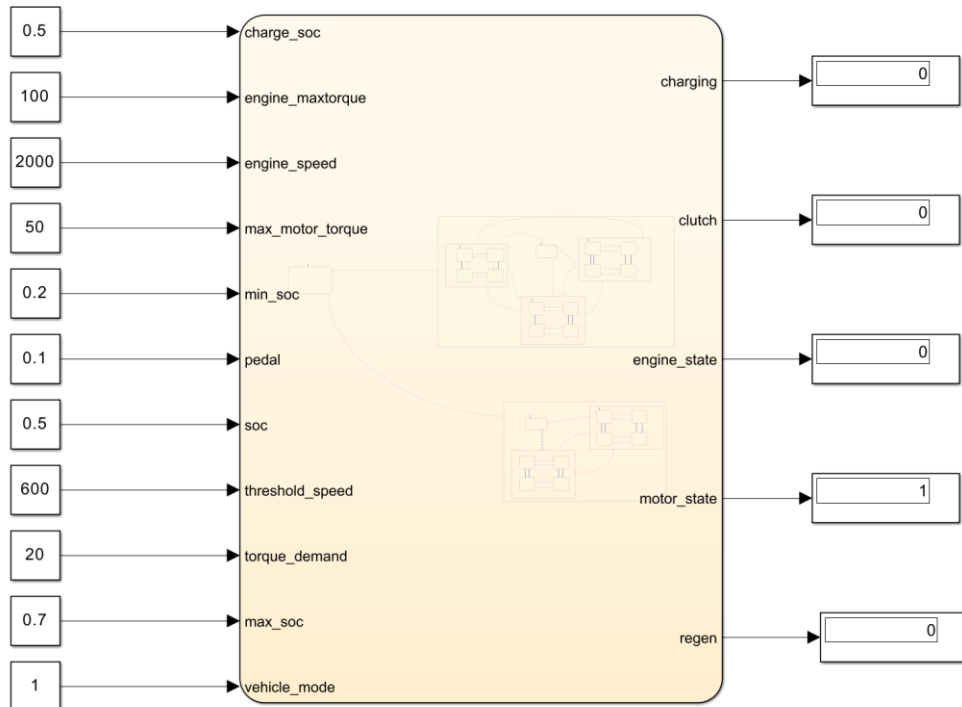


- 1) In sports mode the vehicle first operates in the EV mode till the engine idle RPM speed is reached. After that it operates in parallel mode.
- 2) Based on the SoC it decides whether it should charge the battery through generator and regen.

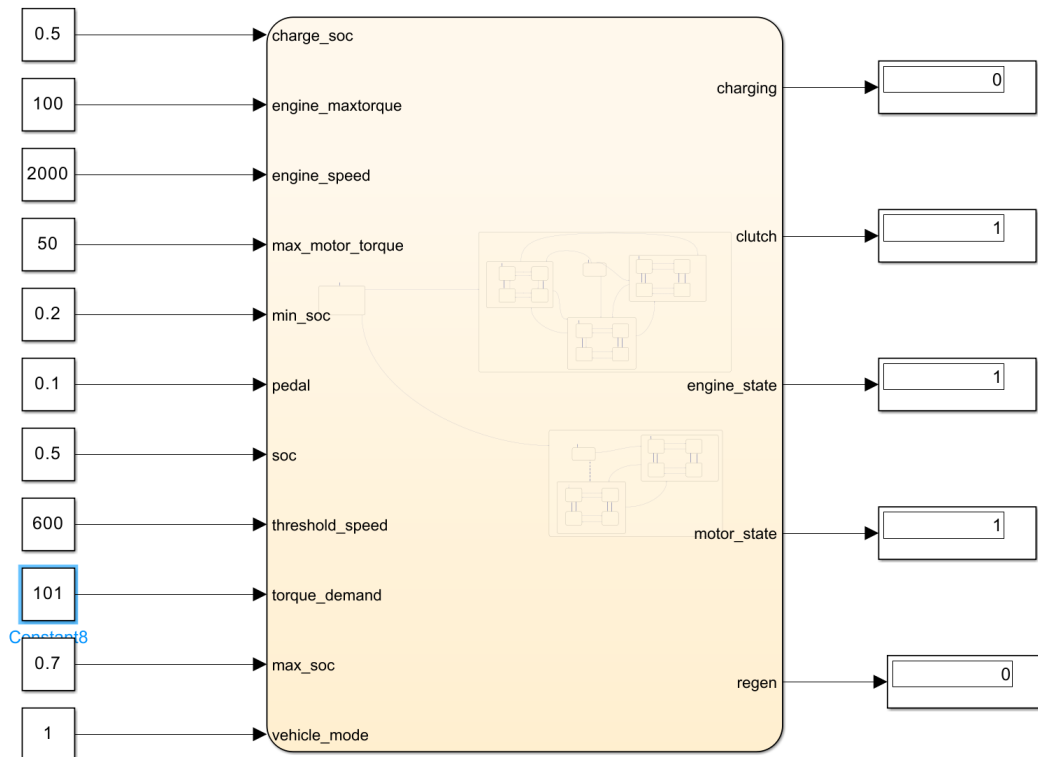
Testing of the control strategy

The control strategy is tested by simulating various conditions of the variables and the corresponding states of the system was observed.

1) EV Mode



2) Parallel mode



RESULTS:

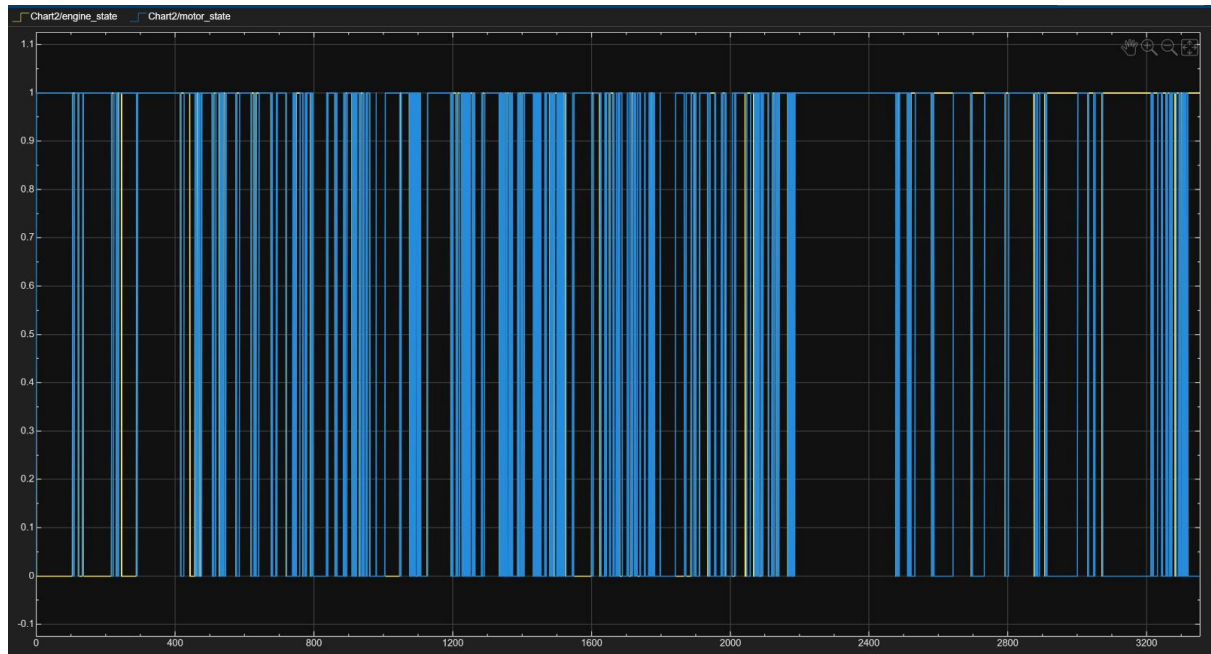


Figure 17 - Engine states and Motor states during the drive cycle

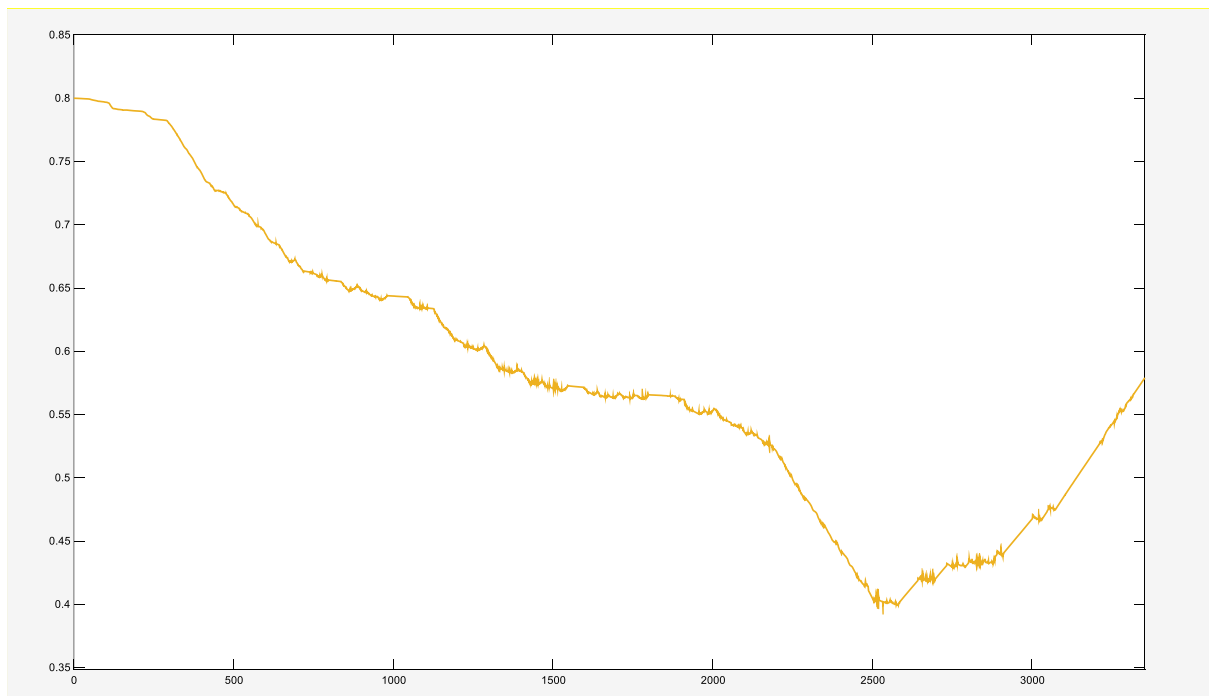


Figure 18: SOC for Hybrid vehicle

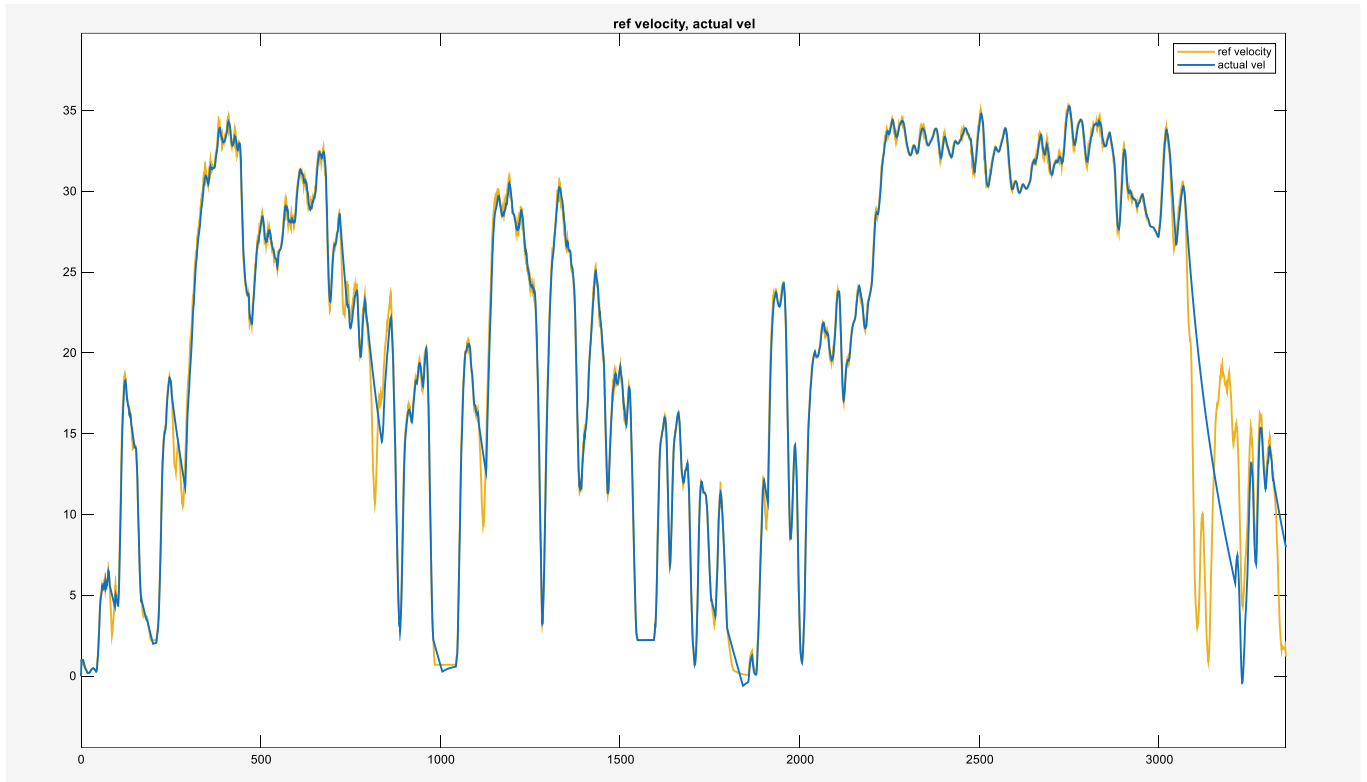


Figure 19: Drive Cycle following with Hybrid powertrain

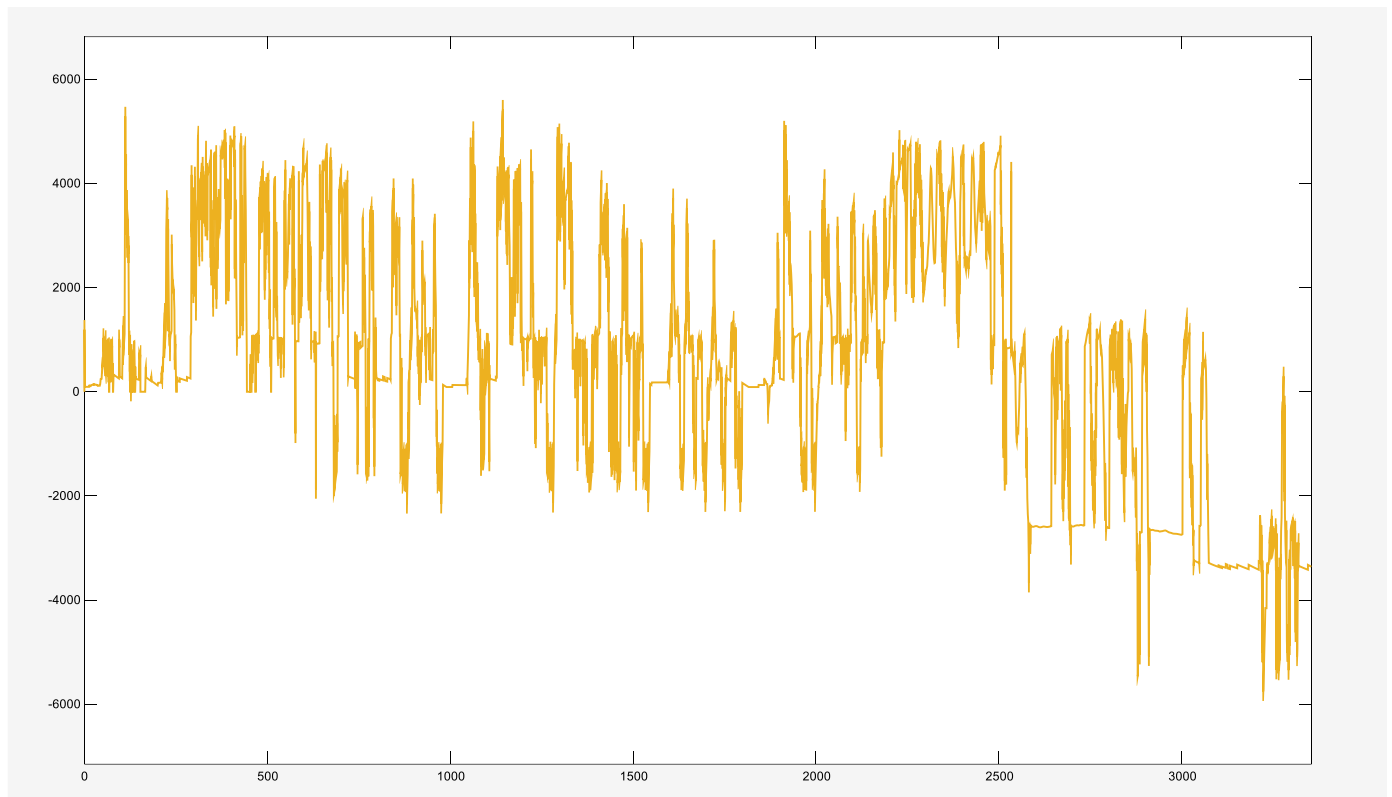


Figure 20: Battery Power for Hybrid vehicle

COMPARATIVE STUDY:

For a battery life of 186472 miles for the EV battery and 93420 miles for the HEV, we compared the purchase and operation costs for the same distance on all three architectures.

Powertrain	Combustion (Toyota corolla 2025)	Electric	Hybrid
Initial Purchase Cost (\$)	22,325	24,725	35000
Fuel/Charing Cost (\$)	11,783	8,391	7283
Battery Replacement (\$)	0	9,000	3000
fuel economy	43 MPG	181.02 Miles per charge	64 MPG
peak power	84.3 kW	82 kW	90
Total Cost (\$)	34,108	42,116	45,283
Range	550 miles (13 gallon tank)	181 Miles/ charge	819 miles (12.8-gallon tank)

CONCLUSION:

After comparing the three powertrain architectures for the same application (passenger/family sedan) along with the comparative cost analysis, the hybrid EV is more efficient in the long run, as it a combination of higher fuel economy and better performance. Keep in mind the cost analysis just factors the battery replacement cost of the HEV assuming it o operates in Pure EV mode. This make the HEV a good option for anyone looking for a high range or range extended option with lower fuel costs.

The series parallel architecture for the hybrid vehicle is run on four main operation modes: motor only, engine only, economy mode and sports mode, where the mode is decided based on the which condition can provide the best driveability (efficiency and power – driver decision) and meet the driver demand. The hybrid architecture is designed around the state-flow control in Simulink, where we were able to achieve the optimal running conditions for the battery to extend the battery life and regeneration abilities. On the other hand, we were successful in fine tuning the powertrain to provide peak performance to the driver as well as best fuel consumption and mileage (64 MPG: 21 MPG more than Conventional ICE Powertrain) while utilising the combustion engine.

Final Recommendation

The series-parallel hybrid powertrain is the optimal choice for our chosen vehicle application, as it delivers best-in-segment performance (90 kW), enormous range (819 miles), and superior fuel economy (64 MPG). Not only does it eliminate range anxiety but also provides value through almost 49% in fuel savings. The only drawback is the total cost of \$45,283 but it makes up for it with higher fuel economy and lower environmental impact. As battery costs decline and manufacturing scales increase, it will make HEVs the most practical and balanced choice for customers seeking a combination of performance and efficiency.

REFERENCE:

App used:

[GPS Logger:](#)

Data conversion:

GPX to XSLX conversion: MyGeoData website: <https://mygeodata.cloud/converter/gpx-to-xlsx>

XLSX to mat: Matlab code

[Engine Data](#)

Motor Data

EV Motor – HVH250

<https://x-engineer.org/ev-design-electric-motor/>

[https://www.evwest.com/support/remy250.pdf?srsId=AfmBOoovKOTcIDZMLtu9y37tGNRjm9t6FHp](https://www.evwest.com/support/remy250.pdf?srsId=AfmBOoovKOTcIDZMLtu9y37tGNRjm9t6FHpF55eRjIR0ZdxJXMK4dA2H)

[F55eRjIR0ZdxJXMK4dA2H](https://www.evwest.com/support/remy250.pdf?srsId=AfmBOoovKOTcIDZMLtu9y37tGNRjm9t6FHpF55eRjIR0ZdxJXMK4dA2H)

Hybrid Motor HPEVS AC-23

<https://www.hpevs.com/hpevs-ac-electric-motors-power-graphs-ac-23.htm>